Curve25519 index calculus: an atlas of coordinates,

summation-polynomial symmetry and factor bases Status 2026-11-04. Written for the owner's question: which coordinates, point representations, equations and factor bases can speed up index calculus on Curve25519, and what each one measured.

Class: a synthesis of measured studies plus the theory that bounds them. Every number has a source path. Nothing here is a leaderboard row. Rho remains ahead on every track, and the distance remaining is stated in each table.

0. How to read this

- **Curves.** The Curve25519 family miniatures built by Curve25519's own rule (`curve25519_family family`): p the largest prime < 2^k with p ≡ 5 (mod 8), smallest A ≡ 2 (mod 4) with #E = 8r, r prime, twist order 4r′. k = 20: p = 1,048,573, A = 926, r = 131,203, √r = 362.2. Also k = 16 (A = 1058), 18 (78), 22 (43186), 24 (318).
- **Units.** Counted only. GAE = group additions equivalent. S = (counted GAE + w × unpriced native operations)/√r, priced at w = 0, 0.1 and 1 GAE per native operation (AGENTS.md §5a). F4 work is counted in multiply-adds and is native (unpriced at w = 0). No wall-clock claim appears anywhere in this atlas.
- **Two tracks.**
  - *Generic track*: a cofactor-projected factor base and a pair table (`prime-projected-abscissa` + `mitm`). This is the lab's baseline, `proj-a`.
  - *Algebraic track*: a factor base with an algebraic description (an x-interval, a tower) and Semaev's summation polynomial solved by F4 (`pkm-semaev`). Coordinates and symmetries only matter here.
- **Cost model** (validated in §6): F4 work per solve E = columns × r × mpt / covered, where columns = base size, covered = exact number of subgroup points that are a sum of two base points, and mpt = F4 multiply-adds per target, which depends on (form, t, G) and not on the seed or the curve.

1. The family's torsion, and what it rules in and out

E(F_p) ≅ Z/8 × Z/r is cyclic for every family curve (the family rule requires it). Consequences:

fact	consequence
E2 = Z/2, generated by T_0 = (0, 0) in Montgomery form	exactly one translation acts on the x-line: u ↦ 1/u. This is the only pairwise symmetry available to summation polynomials over F_p.
rational points of order 4 and 8 exist (u = ±1 have order 4)	translation by them is not a function of x. Any coordinate on which T_4 acts is invariant under $2T_4 = T_0$, so it lives on E/⟨T_0⟩, where T_4 has become a 2-torsion point: the same symmetry on an isogenous curve, not a new one (`research/notes/index-calculus/RESEARCH_EXOTIC_COORDINATES.md` §1.3).
E2 ≠ Z/2 × Z/2	Jacobi intersection and Huff models do not exist over F_p; the full FGHR symmetry group (which needs E[2] rational, or an extension field) is not available.
no rational 3-torsion (#E = 8r)	Hessian and twisted Hessian models do not exist over F_p.
j ≠ 0, 1728	Aut(E) = {±1}: no GLV/GLS orbit folding. The strongest rho is the negation-map walk, `rho-neg`.
prime field	no Frobenius; no Weil restriction gain.
p ≡ 5 (mod 8)	2^t ∤ p − 1 for t ≥ 3: the PKM Kummer tower exists only for t ≤ 2. p − 1 = 4 × 3³ × 7 × 19 × 73 at k = 20, so odd multiplicative subgroups exist (lead L11, §9).
cofactor 8, cyclic	every point has a cofactor coset c ∈ Z/8. A sum of two base points lands in the subgroup iff their cosets cancel. This is the quantity every algebraic-track gain in this atlas comes from (§5).

2. Coordinates and curve models

Every separable degree-2 coordinate f: E ⟶ P¹ is the quotient by an involution P ↦ T′ − P; conjugating by a translation moves T′ to O, so f = N(x(P + S)) for a Möbius map N ∈ PGL_2 and a point S (`src/cryptanalysis/coordinate_search.rs`, checked numerically there). A factor base "f(P) in an interval" is therefore a Möbius frame of x, translated by S. The translation shifts every log by log S; on the algebraic track that changes the coset of every pair sum by 2·c(S) and nothing else.

model / coordinate	exists on the family?	degree on E	relation to x	measured	status
short Weierstrass x	yes	2	identity	frame census; ladder Q-A 458 signed classes at width 2^10 (λ = 0.959); engine `bits` cells	the reference frame, lo = 0 is one draw of the frame distribution

Montgomery u = x − A/3	yes	2	translation of x by a constant	ladder Q-A 533 (0.987)	same frame family, shifted
Edwards y = (u − 1)/(u + 1)	yes	2	Möbius of u	ladder 499 (0.979); engine: same solving degree and terms as `bits` in 8/8 (t, G) pairs; per-target cost equal except 0.6386× at t = 5, G = 0 (T6: real on 7/7 and 3/3 seeds, gone at G ≥ 1 and at t = 4, 6)	a frame; the G = 0 anomaly has no use at the best cells
twisted Edwards x (a = A+2, d = A−2) and Ed25519 x (a = −1)	yes	2	odd under negation, so **not** a function of u: it is the x-line of the translate by a point Q of order 4 (x_Ed is invariant under P ↦ T_0 − P), composed with a Möbius map	smoke census only (projected setting, inside the null)	a translated frame. Derivation: base points are τ_Q of a frame base, so pair sums land in R + T_0 (coset 4), never in the subgroup, unless the oracle also accepts R + T_0 (lifts)
Jacobi quartic	yes (needs rational 2-torsion)	2	Möbius of x	not run; same family by §2	frame
Doche–Icart–Kohel 2-isogeny form	yes	2	Möbius of x	not run	frame
Kummer line, level-2 theta	yes	2	Möbius of x	`prime-kummer` tower exists only for t ≤ 2 (p ≡ 5 mod 8)	frame; the tower membership is unavailable
Tate normal form for the point of order 8	yes	2	Weierstrass shift	not run	frame
Kohel μ_4-normal form	needs rational 4-torsion: yes	2	Möbius of x	not run	frame
Hessian, twisted Hessian	**no** (no 3-torsion)	—	—	—	not defined over F_p
Jacobi intersection, Huff	**no** (need Z/2 × Z/2)	—	—	—	not defined over F_p
binary λ-coordinates	characteristic 2 only	—	—	—	n/a
T_0-invariant w = u + 1/u (Edwards y^2 likewise)	yes	2 on E/⟨T_0⟩	x-line of the 2-isogenous quotient	ladder 268 (0.669) and 245 (0.604); engine: equal solving degree and terms, per-relation 1.24× (t = 5), 1.73× (t = 6), 3.00× (t = 4) dearer than plain at equal base size (`curve25519_t0_algebraic_20261102`); generic track T_0 ÷ blind yield 1.944–2.016, projection ÷ T_0 3.33–4.00 (`curve25519_t0_symmetry_20261102`)	closed; frame selection on it (T7) is 0.93–0.95× the best bits frame per relation, and the isogeny tower is 3.9–79× cheaper
2-descent u = w^2 (even cosets)	yes	4	not a frame	coverage per column² 1.89–2.02×; but solving degree 4 ⟶ 6, 5 ⟶ 7 and F4 per relation 1.28× (t = 4), 1.71× (t = 5) dearer even at T2's oracle-best frame (`curve25519_torsion_leads_20261103`)	closed
y-coordinate, or any degree-3 coordinate	yes	3	not a frame; x is a cubic root	not run	closed by the same argument as 2-descent: a non-Möbius membership raises the

					per-summand degree, and degree 2 already lost

The census result that settles the model question. On five family curves, over 2,000 random translation frames and 2,000 random Möbius frames per (curve, t), Weierstrass lo = 0 and Edwards y are ordinary draws: 49 of 50 cells inside the central 98% (P1), translation ≡ Möbius with mean ratio 0.986–1.012 (P2) (`curve25519_frame_search_20261103`). Edwards "beat" Weierstrass in the earlier study only because lo = 0 was a below-average draw on that curve.

What does work: choosing the frame. The F4 system is identical in every frame. Choosing lo per (curve, t) by the log-free coset count J (read from [r]P) raises coverage 2.5–2.8× (t = 5), about 2× (t = 6), 1.5–1.7× (t = 7) over the mean frame, at unchanged per-target cost; in the engine the F4 work per relation fell 4.71× (t = 5) and 3.33× (t = 6) against lo = 0 at k = 20.

3. Summation polynomials and their symmetry

3.1 The forms in use

- Weierstrass $S_3(x_1, x_2, x_3)$ for $y^2 = x^3 + ax + b$, as a quadratic in x_3: $(x_1 - x_2)^2 x_3^2 - 2[(x_1 + x_2)(x_1x_2 + a) + 2b]\, x_3 + (x_1x_2 - a)^2 - 4b(x_1 + x_2)$ (`pkm_semaev.rs`, `s3_in_u`). S_4 is the resultant of two S_3's.
- Montgomery form in u: the same polynomial after x = u + A/3; its coefficients involve A only. Over F_p coefficient size is irrelevant to F4, and the term structure is the same.
- Edwards y form: S_3 with x_i = (1 + s) + (1 − s) y_i over 1 − y_i, denominators cleared (`Membership::EdwardsBits`). Same degree.
- T_0-quotient form: the ordinary S_3 of $E/\langle T_0\rangle$ at w_i + s equals $S_3(z_1, z_2, z_3) \cdot S_3(d/z_1, z_2, z_3)$ up to a monomial (unit tests `t0_symmetrised_polynomial_is_s3_times_its_t0_image`, and the baseline identity $S_3(1/z_1, 1/z_2, z_3) = S_3/(z_1z_2)^2$). It vanishes exactly when $\pm P_1 \pm P_2 \pm P_3 \in \{O, T_0\}$.

3.2 The symmetry group of the relation on this family

A symmetry of E acts on the x-line iff it normalises [−1], i.e. is a translation by 2-torsion or an automorphism. On the relation $P_1 + \cdots + P_{m+1} = O$:

symmetry	available here?	scope	effect on S_{m+1}
S_m permutations of the summands	always	global	S_{m+1} is symmetric in $x_1 \ldots x_m$; writing it in elementary symmetric functions divides the Gröbner work by about m! only for m ≥ 3
negation	always	per point	already quotiented out by using x
translation by $T_0 \in E2$	yes (one point)	pairwise: applied to an even number of summands	the invariant ring halves the degree in the invariant variable (FGHR); on $E/\langle T_0\rangle$ this is the quotient's own S_3
translation by the other 2-torsion points	no (irrational)	—	—
translation by 4- or 8-torsion	no action on the x-line	—	per-point coordinates cannot see it (§1)
Aut(E)/±1	trivial (j ≠ 0, 1728)	—	no orbit collapse
Frobenius	none (prime field)	—	no Weil restriction

So the group acting is S_m × Z/2, and it is the smallest case in the literature. What it buys, measured:

- At m = 2, S_3 has degree 2 in each variable, and symmetrisation cannot lower it. Measured: the T_0 form has the same solving degree, the same terms and the same F4 cost per target as the plain form in every pair (`curve25519_t0_algebraic_20261102`). Its only effect is on the base: closing the base under $u \mapsto 1/u$ doubles the sums reached (R and R + T_0), but at equal column count the closed base has fewer distinct sums, so per relation it costs 1.2–3.0× more.
- At m = 3, S_4 has degree 4 per variable and symmetrisation would matter more, but m = 3 is 2,297× (t = 4) and 6,643× (t = 5) costlier per relation than selected m = 2 at k = 20 (T3). A factor of a few from symmetry does not close that.
- The pairwise T_0 action on the *target* side (accept relations for R + K, K ∈ E[8], the `lifts` oracle) is the generic-track projection in disguise: on the generic track projection recovers all 8 cosets and beats T_0 keying by 3.3–4.0×; on the algebraic track, after frame or seed selection, lifts add only 1.00–1.12× (T1).

3.3 Why the symmetry story is short on this family

FGHR's degree reduction comes from $E[2] \cong Z/2 \times Z/2$ (or from Frobenius over F_{p^n}), and the big index-calculus gains in the literature come from $m \geq 3$ with those symmetries. Curve25519's rational 2-torsion is $Z/2$ and its field is prime, so the only symmetric reduction available is the T_0 one, and at $m = 2$ it reduces nothing. Everything the algebraic track gained here came from *selection* (frames, seeds, G), not from symmetry.

4. Factor-base families

family	membership	free parameter	F4 system shape	best measured at k = 20	status
`prime-projected-abscissa` + `mitm` (generic baseline `proj-a`)	smallest x, projected by 8	size N	none (pair table)	S = 12.05 / 12.93 / 20.83 at w = 0 / 0.1 / 1 (T5 session), 8.3–8.5× rho-neg there; 7.0–7.2× in `curve25519_algebraic_ecbench_20261102`	the baseline. Coverage sits at the pairs/r ceiling, so no coordinate lever exists here
x-interval, lo = 0 (`bits`)	$x \in [0, 2^t)$	—	t bits per summand, $b^2 = b$	per relation $9.71 \cdot 10^8$ (t = 4), $9.49 \cdot 10^9$ (t = 5), G = 3	the old algebraic reference
x-interval, selected frame (`bits@L`)	$x \in [L, L + 2^t)$	L, chosen by argmax J (log-free)	identical	per relation $1.49 \cdot 10^8$ (t = 4, 6.5× lower), $2.01 \cdot 10^9$ (t = 5, 4.7× lower)	validated (`curve25519_frame_search_20` end to end with T11's `prime-interval` base: 0.437 [0.300, 0.637] vs lo = 0 (validated)
Edwards-y interval	$y \in [0, 2^t)$	(frame)	same	equal to bits except the t = 5, G = 0 anomaly	a frame
T_0-interval (`t0-bits@L`)	$u + 1/u \in [L, L + 2^t)$	L	same degree	selected: 0.951× / 0.934× the best bits frame per relation (t = 4, 5), 1.25× at t = 6	closed (T7)
2-descent (`square-bits@L`)	$x = w^2 + A/3$, $w \in [L, L + 2^t)$	L	degree 4 per summand	1.28× / 1.71× dearer than selected bits	closed (T2)
PKM Kummer tower	$x \in g \cdot \mu_{2^t}$	g	t squaring rules	$t \leq 2$ only here	unavailable
PKM isogeny tower (`prime-isogeny:t,seed`)	$x \in x(R' + \langle Q \rangle)$ on an auxiliary curve, via a Vélu 2-isogeny chain	seed (auxiliary curve and coset)	t tower rules, 16 terms at (t = 6, G = 4)	seed 0: 25 columns, 172 covered; seed 1093 (argmax J): 46 / 528; **seed 1308 (argmax J/ columns): 36 / 498** at t = 5 seed 1307: 22 / 210	the best algebraic family; see §6
odd Kummer $x \in g \cdot \mu_d$, $d \mid (p-1)/4$	multiplicative subgroup of odd order	g, d	odd-degree power rules (not implemented)	—	proposed, design only (L11, wave 4)
digit-set frame $x = lo + \Sigma\, c_j\, b_j$	arbitrary digits c_j	lo, $c_1 \ldots c_t$	identical to bits	—	census running (L14, wave 4)
m = 3 with any of the above	three summands	—	S_4, degree 4 per variable	2,300–6,600× costlier per relation than m = 2	closed (T3)
`explicit-x`, `known-log`	lists / controls	—	—	—	instrumentation

5. Cosets: the mechanism behind every algebraic-track gain

Write $c(P) \in Z/8$ for the cofactor coset of a base point, read off $[r]P$ without any discrete log. A pair $P_i \pm P_j$ is a usable relation for a subgroup target iff $c_i \pm c_j \equiv 0 \pmod 8$.

- For a signed pair of uniformly random cosets, P(c_a ± c_b ≡ 0) = 14/64 ≈ 0.22. The census counts something different: covered subgroup points (±R counted separately) over unordered column pairs (n(n+1)/2). On that denominator a uniform base gives about 0.50 (seed 0: 172/325 = 0.53, seed 1093: 528/1081 = 0.49).
- Selection pushes this up. For seed 1308 at t = 6, 498 covered points over 36·37/2 = 666 column pairs: 0.75, that is 1.5× the uniform value on the same denominator (validator E4).
- The selectors used: J = number of cancelling pairs (frames, seeds); J/columns (T9) because end-to-end cost ∝ columns × r / coverage. J/cols picked the best seed of all 2,000 in 14 of 15 (k, t) cells.
- Filtering a base to a single coset class cannot help: keeping only coset 4 (or only 0) keeps about 1/8 of the columns and makes every pair cancel (at most 2× the selected base's 0.75 on this denominator), which loses to keeping 8× the columns (cost ∝ 1/(columns × p_cancel)).
- Restricting cosets *algebraically* (2-descent, T_0 quotient) raises the polynomial degree and loses; *selecting* among parameterisations with an unchanged system wins. That is the lesson of waves 1–3.
- Torsion-shifted targets (decompose R + T_c, c chosen from the coset histogram) add 1.00–1.12× after selection (T1 failed its 1.10× rule in 19 of 20 cells).

6. End-to-end ledger at k = 20 (full discrete-log solves, ecbench)

Public targets; every run verified by ecbench against the planted scalar. T5 ran 1 measured round (6 runs per arm), T9 and T12 ran 2 (32 per arm). Replay audits: T5 8/8 and T9 16/16 reproduced with 0 problems; T12 one replay per chunk, 16/16 reproduced with 0 problems.

session	targets	arm	S (w=0)	S (w=0.1)	S (w=1)
T5 `ECBS1h405f20219036`	6	rho-neg	1.447	1.547	2.448
		proj-a	12.053	12.931	20.831
		pkm-iso t6 G4 seed 0	29.892	1.674e6	1.674e7
		pkm-iso t6 G4 seed 1093	26.448	1.073e6	1.073e7
T9 (wave 3)	16	pkm-iso t6 G4 seed 1308 ÷ seed 0	0.643 [0.509, 0.813]	**0.5605 [0.421, 0.747]**	0.5605
		seed 1308 ÷ seed 1093	0.723	0.905 [0.740, 1.106]	0.905
T12 (16 one-target chunks)	16	rho-neg	1.544	1.654	2.651
		proj-a	12.633	13.625	22.554
		pkm-iso t6 G4 seed 0	30.906	1.735e6	1.735e7
		pkm-iso t6 G4 seed 1308	16.631	7.816e5	7.816e6
		pkm-iso t5 G4 seed 1307 (T10 winner)	26.404	**1.003e5**	**1.003e6**
L12 (8 chunks, wave 4)	8	winner plain	24.716	9.246e4	9.243e5
		winner + trace replay	24.716	**4.384e4**	4.382e5
L13 (16 chunks, wave 4)	16	winner F4	27.036	1.034e5	1.033e6
		grid on the winner's base	27.036	2,462.7	24,383.6
		pair table on the winner's base	28.310	32.33	68.50
T13 (16 chunks, wave 4)	16	winner (t5 G4 s1307)	28.491	1.106e5	1.105e6
		pkm-iso t4 G3 seed 1193	44.640	6.616e4	6.612e5
		t4 G3 seed 1193 + trace replay	44.640	**3.136e4**	**3.132e5**

Wave-4 paired ratios (pooled, 95 % intervals): L12 trace ÷ plain = 0.4742 [0.47418, 0.47425] at w = 0.1, exactly 1 at w = 0. T13 t4 ÷ winner = 0.598 [0.509, 0.704] at w = 0.1 and 1.567 at w = 0; t4-trace ÷ t4 = 0.4740; **t4-trace ÷ winner = 0.2837 [0.2413, 0.3335] at w ≥ 0.1**. L13 grid ÷ F4 = 0.0238; pair table ÷ grid = 0.0131 (w = 0.1).

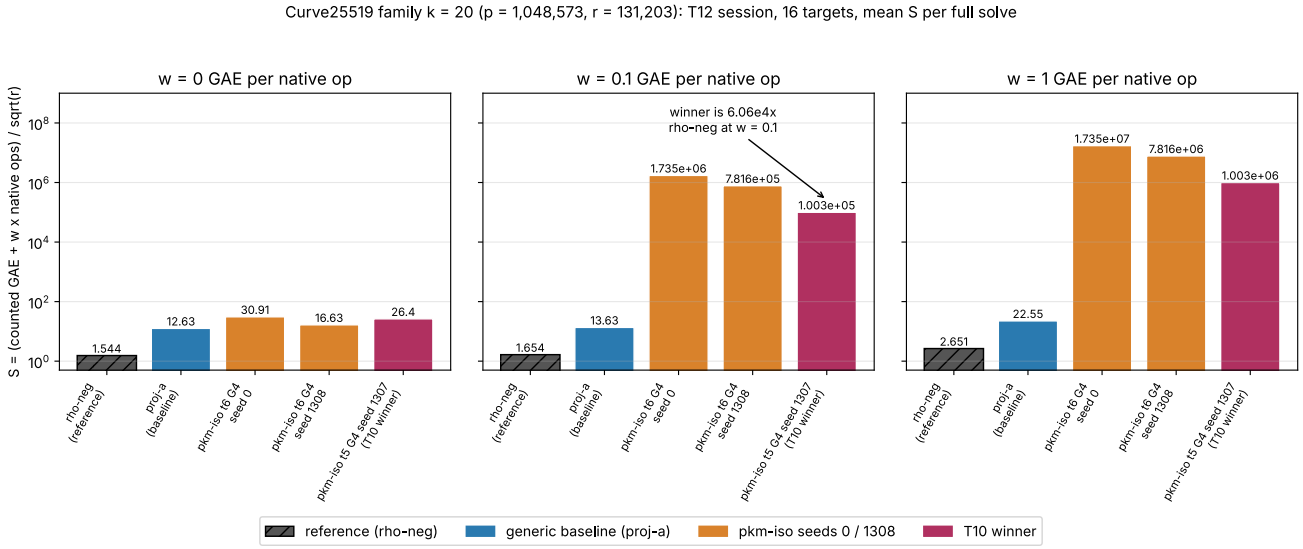
Paired ratios in T12 (K = 16, linearised t): winner ÷ seed 0 = 0.0578 [0.0466, 0.0716] at w ≥ 0.1 (cheaper on 16/16 targets); winner ÷ seed 1308 = 0.128 [0.097, 0.169] at w ≥ 0.1 but 1.588 [1.289, 1.955] at w = 0 (the predicted order flip: the smaller t = 5 base needs more trials per relation, 22/210 against 36/498).

Model against measurement. The cost model predicted winner ÷ seed 0 = 0.054 (measured 0.0578), winner ÷ seed 1308 = 0.108 (0.128), and 1.45 at w = 0 (1.59). T9's model 0.497 against 0.5605. The model is good to 10–20%.

Search costs, per curve. Seed search at t = 6: $1.35 \cdot 10^7$ GAE ($3.7 \cdot 10^4$ in S); at t = 5: $6.94 \cdot 10^6$ GAE ($1.92 \cdot 10^4$ in S). Charged to a single target the searches lose at w = 0 and still win at w ≥ 0.1; amortised over targets on the same curve they are negligible.

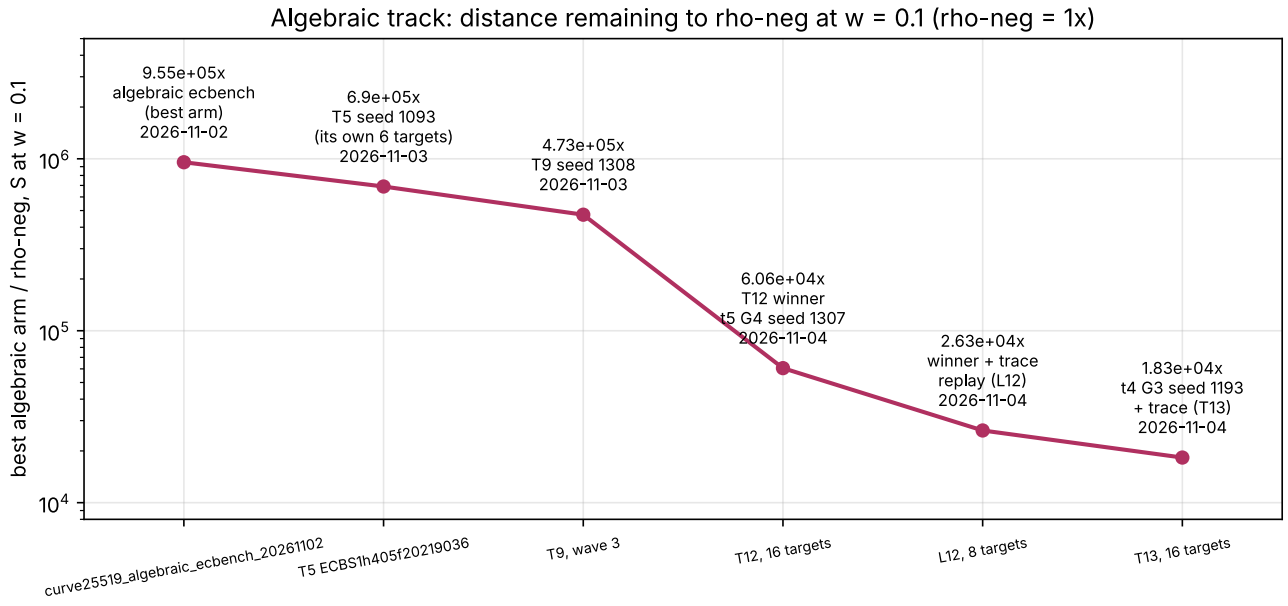
Distance remaining, best F4 arm at w ≥ 0.1 (T13 targets): t = 4, G = 3, seed 1193 with trace replay is 27.9× behind rho-neg at w = 0, **$1.83{\cdot}10^{4}$× [$1.55{\cdot}10^{4}$, $2.15{\cdot}10^{4}$] at w = 0.1**, $1.13{\cdot}10^{5}$× at w = 1. At w = 0 the T12 winner stays the better algebraic arm (17.8× behind rho-neg on T13's targets).

Distance remaining (T12 targets, before wave 4): the best algebraic arm is 17.1× behind rho-neg at w = 0, **$6.06{\cdot}10^{4}$× at w = 0.1**, $3.78{\cdot}10^{5}$× at w = 1; and 2.09× / $7.36{\cdot}10^{3}$× / $4.45{\cdot}10^{4}$× behind the generic baseline. For comparison, on their own target sets: the 2026-11-02 best algebraic arm was $9.55{\cdot}10^{5}$× behind rho-neg at w = 0.1 (`curve25519_algebraic_ecbench_20261102`), T5's seed 1093 $6.9{\cdot}10^{5}$×, and T9's seed 1308 $4.73{\cdot}10^{5}$× (`wave3/t9_seed_per_column/README.md`). The track moved about 16× at w = 0.1 in two days, from selecting seed, t and G. Target sets differ between sessions, so these are descriptive, not paired.



Source: research/curve25519_t10_winner_ecbench_20261104/ecbench/band_pooled.json (T12, 2 rounds, 32 verified runs per arm); REPORT.md section 6.

Figure 1: Mean S at w = 0, 0.1, 1 for the section 6 ledger arms, T12 session (log scale). Source: band_pooled.json.



Source: REPORT.md sections 6 and 11; pooled_compare.jsonl (T12, L12, T13), curve25519_isogeny_seed_20261103/ecbench/band.json (T5).

Figure 2: Best algebraic arm against rho-neg at w = 0.1 over time; target sets differ between points. Source: section 6.

7. The grid limit, stated plainly

The winner (t = 5, G = 4) solves 16 tiny F4 systems per target, 52,816 multiply-adds in total. The exhaustive engine on the same base evaluates S_3 on every pair of the 32 x-values of V: about $n^2 + 7n$ = 1,248 F_p multiplications per target from the grid code. So the best F4 arm is still roughly 40× above brute force on its own factor base, and the generic pair table (`proj-a`) is $7.4{\cdot}10^{3}$× ahead of it at w = 0.1. The joint (t, G, seed) optimisation has moved the F4 track *toward* the grid limit, not past it. Lead L13

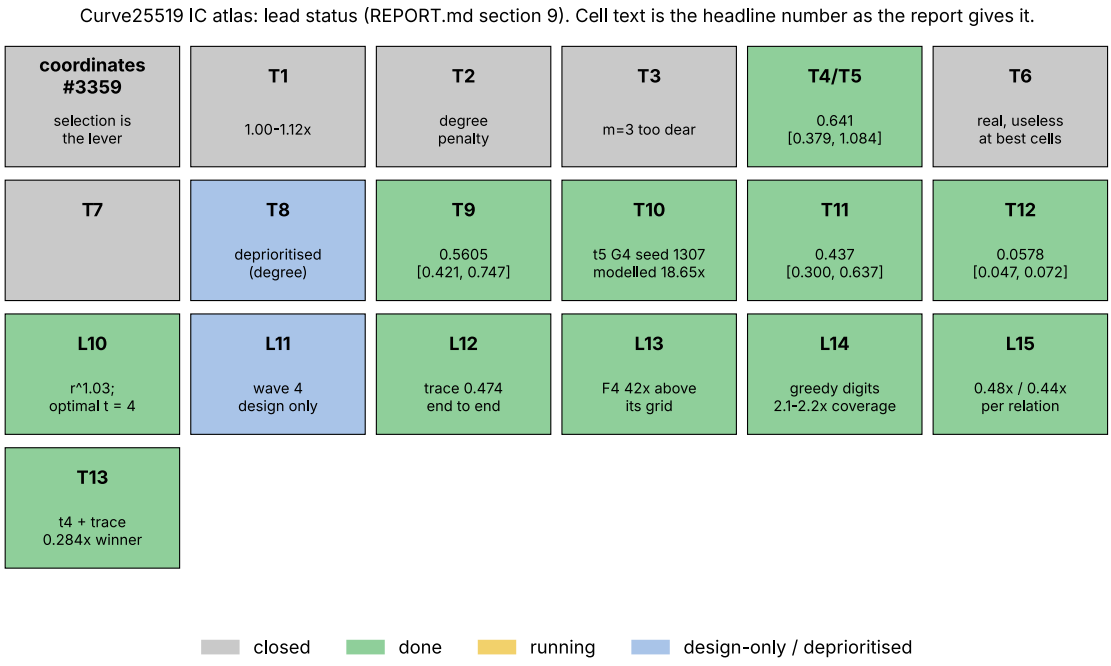
(wave 4) measured this end to end with `engine=exhaustive` and `mitm` on the winner's base: grid ÷ F4 = 0.0238 at w = 0.1, so the F4 arm is 42× above its grid. The pair table is another 76× below the grid, and is still 2.4–3.3× behind proj-a at every weight. On this base no oracle reaches proj-a; the 22-column base sets the floor (§11.3). Until the algebraic track beats its own grid, its value is in how it scales, not in its constant.

8. Scaling

At fixed (t, G), mpt does not depend on k (T4-rep, T6, T10), and the *distribution* of columns and covered over seeds at fixed t does not either (the selected values vary by curve, 28/198 … 39/364 at t = 6), so E = columns × r × mpt / covered is linear in r up to that spread, against rho's √r. The algebraic track at m = 2 can only beat exponent 1 if the optimal t grows with k and mpt(t, G) grows slower than coverage. At k = 20 the optimal t is 5 and mpt jumps 13× from (5, 4) to (6, 4). Lead L10 (wave 4) measures the optimum at k = 16, 18, 22, 24 and fits the exponent. No 255-bit statement is made or implied.

9. Leads: status and what remains

id	lead	status
coordinates (#3359)	model choice	closed: all frames; selection is the lever
T1	torsion-shifted targets	closed (1.00–1.12×)
T2	2-descent base	closed (degree penalty)
T3	m = 3 selection	closed (selection works, m = 3 too dear)
T4/T5	seed by J, end-to-end	done: 0.641 [0.379, 1.084]
T6	Edwards G = 0 anomaly	real, useless at best cells
T7	T_0-base frames	closed
T8	4E/8E halving filters	deprioritised (degree)
T9	seed by J/columns, end-to-end	done: 0.5605 [0.421, 0.747]
T10	joint (t, G, seed)	done: t = 5, G = 4, seed 1307, modelled 18.65×
T11	`prime-interval` base in ecbench	done: selected frame 0.437 [0.300, 0.637] vs lo = 0; 7.71× behind iso seed 1093 (validated)
T12	T10 winner end-to-end	done: 0.0578 [0.047, 0.072] vs seed 0
L10	scaling exponent across k	done: r^1.03, optimal t = 4 (validated)
L11	odd Kummer base μ_d	wave 4, design only
L12	F4 trace replay on the winner	done: 0.474 end to end (validated)
L13	grid-limit control	done: F4 42× above grid at w ≥ 0.1; pair table 2.4–3.3× behind proj-a; P4, P8 failed (validated)
L14	digit-set frames	done: greedy 2.1–2.2× coverage; L15 engine 0.48×/0.44× per relation (validated)
T13	t = 4 candidate + trace end to end	done: 0.284× the T12 winner at w ≥ 0.1 (validated)



Source: REPORT.md section 9 (status 2026-11-04).

Figure 3: Lead status grid from section 9.

Ideas considered and not pursued, with the reason:

- **Two towers with complementary cosets** (cross pairs only): bounded by 1/0.75 ≈ 1.33× over the selected single tower, below the measurement noise of a 16-target session.
- **Montgomery-form S_3 per-target cost**: coefficient size is irrelevant mod p and the term structure is identical; the only measured form-dependence (Edwards, G = 0) vanishes at the cells that matter.
- **Edwards-x bases**: a translate by an order-4 point; identical to a frame plus lifts.
- **y-coordinate or higher-degree coordinates**: degree penalty, as T2.
- **Working on $E/\langle T_8 \rangle$ or another isogenous curve**: the discrete logarithm is isogeny-invariant and the isogenous curve has its own cofactor cosets; nothing is gained beyond moving the frame.
- **Weil restriction to F_{p²}**: the target and base stay in E(F_p); no new symmetry appears.
- **Coset-pure filtering**: loses to selection (§5).
- **Multi-target amortisation** (precompute base logs once): a reporting convention (`ecbench amortize`), not a new method; it does not change the per-relation F4 cost that dominates.

10. Validation state

- Independently validated (`validator-report.json` in each directory): frame search (disposition `validated`, after an `errata_open` pass whose E1 reran the census on the family-rule curves), wave 2 (T3, T6, T7, T4-rep) and wave 3 (T9, T10), all pass_with_errata with errata applied.
- Not independently validated: the torsion leads (T1–T3 census) and the isogeny-seed study (T4/T5). Their census instruments are the frame study's, and T4-rep later reproduced the T4/T5 engine cells exactly, but no validator report exists for them.
- This atlas: `validator-report.json`, pass_with_errata (E1–E8 applied).
- Validated after the first draft: T11 (post-data), T12, L10, L12 (both stages), L14, L15, all pass_with_errata with errata applied.
- Validated last: T13, pass_with_errata. No study in this atlas is pending validation.
- No study here passes the leaderboard validation gate: component results have no same-unit rho row, by design.

Sources

- `research/curve25519_frame_search_20261103/` (frames; P1–P5)
- `research/curve25519_torsion_leads_20261103/` (T1–T3, LEADS.md)
- `research/curve25519_isogeny_seed_20261103/` (T4/T5; `ecbench/s1`)
- `research/curve25519_leads_wave2_20261103/` (T3, T6, T7, T4-rep)
- `research/curve25519_leads_wave3_20261103/` (T9, T10, T11)
- `research/curve25519_t10_winner_ecbench_20261104/` (T12)
- `research/curve25519_algebraic_ladder_20261102/` (Q-A support, Q-D cost)
- `research/curve25519_t0_algebraic_20261102/`, `curve25519_t0_symmetry_20261102/`

- `research/curve25519_edwards_semaev_20261102/`, `curve25519_algebraic_ecbench_20261102/`
- `research/notes/index-calculus/RESEARCH_EXOTIC_COORDINATES.md`, `src/cryptanalysis/coordinate_search.rs`
- `src/cryptanalysis/ic_framework/pkm_semaev.rs` (forms, T_0 identity tests, grid engine)

11. Wave 4 (2026-11-04): scaling, trace replay, grid limit, digit

sets, odd Kummer Protocols frozen together at `5a800bb3d0` before any data (`research/curve25519_leads_wave4_20261104`). Status at the time of writing: L10, L12 (both stages), L13, L14, L15 and T13 independently validated, pass_with_errata with errata applied; L11 is design only.

11.1 L10: the exponent (validated)

40 engine cells, $k \in \{16, 18, 20, 22, 24\} \times t \in \{4, 5, 6, 7\}$ at $G = t - 1$ with the J/columns seed, plus controls. All seven predictions held.

- **mpt is k-invariant to the digit** in every hit-free cell: 17,592 (t = 4, G = 3), 52,816 (5, 4), 146,464 (6, 5), about $4.2 \cdot 10^5$ (7, 6). Seed changes it by < 2%.
- **The optimal t is 4 at every k**, with t = 5 second at 1.7–2.5×. T10 had scanned $t \geq 5$ only, so its winner (t = 5, G = 4) is not the optimum. The next end-to-end candidate is **t = 4, G = 3** with the J/columns seed (1193 at k = 20: 8 columns, 52 covered), modelled 2.04× below T12's winner.
- **E_best(k) ∝ r^1.03** (frozen analysis 1.0297; 1.0315 with the validator's per-target native-work correction). Per step in t the ratio columns × mpt / covered grows 1.6–2.5×, so beating exponent 1 would need coverage to grow 8–19× per step against the measured 2–5×.
- **The gap to rho grows as r^0.53**: in the F4-phase model, $1.5 \cdot 10^4$× (k = 16) to $2.8 \cdot 10^5$× (k = 24) at w = 0.1, with rho's S fixed at 1.5 (a conservative bound; T12 measured 1.65 at w = 0.1).
- Errata: the protocol carried $r_{18} = 32{,}719$ instead of 32,887 (0.5%, no verdict changes); the frozen script priced native work per 100 targets instead of per target (b moves 1.0297 ⟶ 1.0315).

This is the atlas's main structural result: on this family, m = 2 algebraic index calculus with F4 scales *worse* than rho by a factor that itself grows like √r, whatever frame, seed or guess depth is chosen.

11.2 L12: F4 trace replay (validated, both stages)

Learning one refutation trace per branch system and replaying it per target cuts the winner's per-target F4 work **0.4717**× (24,912 against 52,816 mul-adds) with identical verdicts, zero fallbacks, and a learning cost of exactly one plain target. At (t = 6, G = 4) the ratio is 0.454 for both seeds; at (5, 3) it is 0.566, so the frozen "≤ 0.5 in every cell" rule failed there. End to end (8 targets × 2 rounds, four arms), all four predictions held: S(trace)/S(plain) = **0.4742 [0.47418, 0.47425]** at w = 0.1 and 0.4741 at w = 1, with the walk, relations and recovered scalars identical (w = 0 ratio exactly 1). Charging the unpriced learning by hand moves it to 0.4744. Distance to rho-neg at w = 0.1: $2.63 \cdot 10^4$× [$1.52 \cdot 10^4$, $4.54 \cdot 10^4$]. Of $1.6 \cdot 10^6$ replays, 223 fell back to plain F4 (59 genuine replay failures). Counted units only: solver wall time is 0.95×, not 0.47×.

11.3 L13: the grid-limit control (validated)

Five arms on the winner's own base, 16 one-target chunks, 32 runs each, all verified. 6 of 8 predictions held.

- **grid ÷ F4 = 0.0238 [0.02382, 0.02384]** at w = 0.1 (frozen point 0.0245), with identical trials, relations and GAE in all 32 pairs. The best F4 arm is 42× above its own brute-force limit.
- Pair table ÷ grid = 0.0131 at w = 0.1 and 0.0028 at w = 1; 1.047 at w = 0 (its 506 table additions).
- **Even the pair table on this base loses to proj-a at every weight: 2.39× / 2.54× / 3.32×** at w = 0 / 0.1 / 1. With the oracle held fixed, the 22-column isogeny base loses to the 32-column projected base. The grid is 2.28× / 194× / 1,181× behind proj-a.
- P4 failed on its per-pair clause: GAE(mitm) − GAE(grid) is 437–489, not 506 ± 2 × relations, because the grid and F4 arms pay about 4 extra lift additions per relation that the table's signed probe does not. P8 failed as the frozen-rule defect: the grid count is exactly 1,248 × solver calls in 32/32 runs, not 1,248 × trials. Neither is retuned.

The consequence for the algebraic track: none of the three m = 2 oracles on this base (t = 5, seed 1307, k = 20) overtakes proj-a, and no oracle can go below the w = 0 row, already 2.3× behind. The preregistered model puts the cause in the base's coverage per column (measured trials per relation 10.1× proj-a's); this run does not isolate it. Validated pass_with_errata.

11.4 L14: digit-set frames (census validated; lead open)

$x = lo + \Sigma\, c_j\, b_j$ with arbitrary digits c_j keeps the F4 system identical to `bits`. Exact census at k = 20:

- Random digit sets and random translation frames have the same coverage distribution (mean ratios 0.985–1.015; the max-ratio rule failed at t = 5 on one outlying translation frame).
- **A greedy digit choice (B = 2,000 candidates per digit, selected by J) covers 522 points on 22 columns at t = 5 and 1,268 on 43 at t = 6, against 248 and 570 for the best of 2,000 translation frames: 2.1× and 2.2×** (rule ≥ 1.3×; the point prediction of 1.1–1.45× failed on the high side).
- Mechanism (exploratory): the greedy builds an additive tower inside a single cofactor coset (16 of 22 points in coset 0 at t = 5), so covered/column is 2.3–2.5× the best translation frame. Replicates spread 1.19–1.49× depending on the starting lo.

- Where it lands: the bits form's per-target F4 cost is 17–64× the isogeny form's at equal (t, G), so a digit-set bits arm models about 6.9× behind the T10 winner at (5, 3).
- **L15 (engine, all five predictions held):** with `Membership::Digits` (`b847203692`) the greedy sets keep the `bits` system exactly (same degree and terms, per-target cost within 1–3%) and deliver their census coverage, so F4 work per relation falls 0.48× (t = 5) and 0.44× (t = 6) against the best translation frame. At matched (t = 5, G = 4) the digit set stays 10.5× behind the isogeny winner (per-target 1,375,663 against 52,816 mul-adds).

11.5 L11: odd Kummer base (design only)

$p - 1 = 4 \times 3^3 \times 7 \times 19 \times 73$ at k = 20 gives multiplicative subgroups μ_d of odd order. A base $x \in g \cdot \mu_d$ needs odd-degree power rules the tower engine lacks. The design specifies the census to run first and a go/no-go rule before any engine work; nothing was measured.

11.6 What wave 4 changes in the lead list

- **T13 (validated):** t = 4, G = 3, seed 1193 at k = 20, all six predictions held. Plain t = 4 is 0.598 [0.509, 0.704] the T12 winner at $w \geq 0.1$, against the model's 0.49. With trace replay (0.4740) it is **0.2837 [0.2413, 0.3335]**, the best F4 arm so far at $w \geq 0.1$: $1.83 \cdot 10^4$× behind rho-neg at w = 0.1. At w = 0 it is 1.567× dearer than the winner, as predicted, so the band leaves the pair unresolved. Counted units only: solver wall time is 0.869×. Validated pass_with_errata.
- **L15:** done (above); at (t, G) = (5, 3) the digit set is 12.4× below the `lo = 0` bits control per relation (derived) and 6.2× per solve, still 10.5× behind the tower (validated pass_with_errata).
- The exponent result bounds all of this: constant-factor gains on a track whose exponent is 1.03 against rho's 0.5.